INCEPTION 2026 Paper and Poster Competition

Integrated Digital Framework for Offshore Facilities: Synergizing UAV-Based Thermal Imaging and Multivariate Transformer Analytics for Early Gas Flare Prediction and Emission Control

[1]Faiz Iqbal I'tishom, [2]Nathanael Indra Raharjo Paseran, [3]Moch Shultan Ali Saifuddin.
[1]Data Science technology, [2]Robotics and AI Engineering, [3]Robotics and Artificial Intelligence Engineering
[1]faiz.iqbal.itishom-2023@ftmm.unair.ac.id, [2]nathanael.in.raharjo-2023@ftmm.unair.ac.id, [3]moch.ultan.ali-2023@ftmm.unair.ac.id.

Abstract

Offshore oil and gas facilities face critical challenges in mitigating greenhouse gas emissions and ensuring operational safety during gas flaring. Conventional monitoring relies on reactive threshold alarms and manual inspections, which are constrained by safety risks and limited spatial coverage. To address these, we propose an integrated digitalization framework combining predictive analytics and external realtime aerial inspections using Unmanned Aerial Vehicles (UAVs).

The framework's first component is an early-warning predictive system powered by a Multivariate Transformer architecture. By analyzing complex, non-linear time-series data from internal gas plant sensors such as pressure, temperature, and flow rate the model utilizes Time Absolute Position Encoding (tAPE) to capture temporal dependencies efficiently. This identifies process anomalies and predicts potential flaring events with a significant Detection Time Advance before flare ratios reach thresholds, granting operators crucial time for proactive intervention.

Complementing the predictive layer, autonomous UAVs equipped with optical gas imaging (OGI) and thermal infrared sensors monitor external conditions. The captured thermal and multispectral imagery is processed through a deep learning segmentation pipeline. Utilizing architectures like U-Net, the model effectively distinguishes flare signatures from other high-temperature anomalies. This computer vision module performs three core functions: precise gas leak localization, flaring safety assessment through smoke opacity classification, and combustion error detection to evaluate burning efficiency.

By synthesizing physics-informed data-driven predictive models with high-resolution UAV thermal imaging, this dual-layer framework offers a highly scalable solution for offshore operations. Preliminary analyses indicate this approach effectively predicts flare precursors and classifies combustion efficiency states with over 90% accuracy. Ultimately, this system minimizes unplanned flaring events, enhances environmental compliance, and significantly reduces personnel exposure to hazardous zones, supporting a sustainable energy transition.

Keywords: UAV inspection, semantic segmentation, early flare prediction, deep learning, offshore maintenance.